Understanding DVA-C02 Domains and Exam Guide

Explore the four essential domains of the AWS Certified Developer - Associate (DVA-C02) exam: development, security, deployment, and troubleshooting. Learn how these areas reflect real-world AWS application life cycles, including coding with SDKs, securing cloud apps, modern deployment practices, and performance optimization to help you prepare confidently for the exam.

Welcome to the starting line for mastering the AWS Certified Developer – Associate (DVA-C02) exam! If you've built an app on AWS or written a Lambda function that solved a real problem, you're already on your way. This certification validates a developer's ability to build modern cloud applications that are secure, scalable, and production-ready.

The exam is structured around four core domains that mirror the real-world life cycle of an AWS-powered application: development, security, deployment, and troubleshooting.

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Exam domains

The exam domains represent the key skill areas AWS expects every cloud developer to master. Each domain corresponds to real-world responsibilities we'll encounter when building and maintaining applications on AWS. Let's take a quick look at the types of tasks and knowledge each domain covers before we begin to explore further:

- **Development with AWS services:** This domain focuses on building scalable, resilient cloud-native applications. We'll need to be familiar with how to:
  - Write code that interacts with AWS services using SDKs and APIs.
  - Design fault-tolerant apps with retries, queues, and dead-letter mechanisms.
  - Implement Lambda functions, including configuration, testing, and integration with other services.
  - Use appropriate data stores—DynamoDB, RDS, or S3—based on access patterns and data types.
- **Security:** This domain emphasizes securing apps and data in the cloud. We should be comfortable with tasks such as:
  - Use Amazon Cognito, IAM, and STS for user authentication and authorization.
  - Apply least privilege policies and federated identities.
  - Use AWS KMS and encryption strategies for data at rest and in transit.
  - Secure secrets using AWS Secrets Manager and Parameter Store.
- **Deployment:** This covers modern CI/CD practices and app packaging. We are expected to:
  - Ensure all Lambda artifacts use current, supported runtimes such as Python 3.9+, Node.js 20+, Java 11/17, or .NET 8+ to maintain compatibility and support throughout the exam and real-world usage.

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  - Prepare deployment artifacts using Lambda layers, container images, or zipped code.
  - Test apps using mocked endpoints, development stages, and automated scripts.
  - Automate deployments using CodePipeline, CodeBuild, SAM, or CloudFormation.
  - Implement deployment strategies like blue/green or canary releases.
- **Troubleshooting and optimization:** This domain emphasizes diagnosing and improving application performance. We are expected to:
  - Analyze logs and metrics from CloudWatch and X-Ray to find bottlenecks or failures.
  - Use distributed tracing and custom metrics for observability.
  - Optimize performance using caching strategies and concurrency settings in services like Lambda, SQS, and ElastiCache.

The AWS Certified Developer – Associate exam guide consists of four domains. The table below shows the domains and the percentage of scored questions in each domain:

Domain	Weight
Development with AWS services	32%
Security	26%
Deployment	24%
Troubleshooting and optimization	18%

We won't be asked to design complete system architectures, administer networks, operating systems, or manage IAM groups. Designing CI/CD pipelines

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from scratch is also outside the expected scope. Our role is focused on using and contributing to existing ones.

Exam guide

Understanding how the exam is structured helps us better prepare and manage our time during the test. The exam consists of 65 questions: 50 scored and 15 unscored (used for future test development). You'll have 130 minutes to complete it. The question types are:

- Multiple choice (*one correct response among four)*.
- Multiple response (*two or more correct answers from five or more choices*).

The exam is pass/fail, with a scaled score range from 100 to 1,000. We need a minimum of 720 to pass. AWS uses a compensatory scoring model, meaning we don't need to pass each domain individually; strong performance in one area can offset weaker performance in another.

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